a5d86fab-0d72-4ad0-a935-cbebff4a5329

<prev_user>

<request>

You are working on an issue in the 'dharmab/skyeye' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/skyeye/skyeye (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-04-22T18:19:04.440+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>Nil dereference in FindNearestGroupWithBullseye</issue_title>
<issue_description>## Summary

In `pkg/radar/nearest.go`, `FindNearestGroupWithBullseye` calls `findGroupForAircraft(nearestTrackfile)` which can return nil when no group is found. The code then unconditionally dereferences the result by calling `grp.point()`, which will panic.

The sibling function `FindNearestGroupWithBRAA` correctly handles this case by checking for nil before dereferencing.

Location

`pkg/radar/nearest.go:99-113`

Suggested Fix

```
    Add a nil check on `grp` before dereferencing, matching the
    pattern used in `FindNearestGroupWithBRAA`.

    🤖 Generated with <a href="https://claude.com/claude-
    code">Claude Code</a></issue_description>

    ## Comments on the Issue (you are @copilot in this section)

    <comments>
    </comments>
    </problem_statement>


  Implement the necessary changes to the repository so that the
  requirements specified in the <problem_statement> are met.
```

<prev_assistant>

Implemented a minimal fix for the nil dereference in `FindNearestGroupWithBullseye`.

✅ Changes made

1. **Nil guard added**
   - **File:** `/home/runner/work/skyeye/skyeye/pkg/radar/nearest.go`
   - **Change:** After:
   ```go
   grp := r.findGroupForAircraft(nearestTrackfile)
   ```
   added:
   ```go
   if grp == nil {
       return nil
   }
   ```
   This prevents dereferencing `grp` when no group is found.

2. **Focused regression test added**
   - **File:** `/home/runner/work/skyeye/skyeye/pkg/radar/nearest_test.go`
   - **Test:** `TestFindNearestGroupWithBullseyeReturnsNilWhenNoTrackfile`
   - Verifies the function returns `nil` (and does not panic) when no trackfile/group exists.

```
---

### Validation run

- Ran baseline commands before changes:
  - `make lint`, `make skyeye`, `make test` paths were attempted.
  - In this environment they fail due to missing system libs (`opus`,
`soxr`) via pkg-config (pre-existing infra/dependency issue).
- Ran targeted test invocation via:
  - `make test TEST_FLAGS='-run
TestFindNearestGroupWithBullseyeReturnsNilWhenNoTrackfile'`
  - Still blocked by the same repo-wide native dependency build
failures.
- Ran final automated validation:
  - **Code Review:** no comments
  - **CodeQL:** 0 alerts

No unrelated code was modified.
```

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'dharmab/skyeye' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/skyeye/skyeye (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-05-05T02:15:12.474+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Guard `FindNearestGroupWithBullseye` against nil group dereference

</pr_title>

<pr_description>
`FindNearestGroupWithBullseye` could panic by dereferencing a nil group when `findGroupForAircraft` returned nil. This aligns bullseye behavior with existing BRAA handling for no-group cases.

- **Bug fix: nil-safe bullseye nearest-group path**
  - Added an early return in `pkg/radar/nearest.go` when `grp == nil` in `FindNearestGroupWithBullseye`.
  - Prevents panics in empty/no-match scenarios and preserves the function contract of returning `nil` when no group is found.

- **Regression coverage**
  - Added `pkg/radar/nearest_test.go` with a focused test:
    - `TestFindNearestGroupWithBullseyeReturnsNilWhenNoTrackfile`
  - Verifies no contacts/no trackfile path returns `nil` instead of dereferencing.

```go
grp := r.findGroupForAircraft(nearestTrackfile)
if grp == nil {
    return nil
}
```
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>4376088367</comment_id>
<author>@dharmab</author>
@copilot resolve the merge conflicts in this pull request
</comment_new>

</pr_comments>

</comments>

The last **2** git commits in this branch are the changes you

```
have made so far. Use those as your change commit history.
</problem_statement>
```

```
Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the
<comments> before making any changes. You will only address the
<comment_new> comments. Reply to new comments as you address them.
Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a
suggestion.
    - Is it addressed to you or someone else, e.g. "Hey
@username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can
stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and
ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>
```